

KRISHNA DESAI

Bloomington, IN | +1(930) 904-5193 | desaikri@iu.edu

<https://github.com/krishnadesai03> | <https://www.linkedin.com/in/krishna-desai-7162b8224/>

EDUCATION

Indiana University Bloomington

Master of Science in Data Science (GPA: 3.7/4.0)

Semester 3 Coursework: Applied Algorithms, Information Visualization, Data Visualization

August 2024 - May 2026

Bloomington, IN, US

University of Mumbai

Bachelor of Engineering in Information Technology (CGPI: 9.52/10.0)

January 2021 – July 2024

Mumbai, MH, India

TECHNICAL SKILLS

Programming Languages: Python 3.10.9, R, C/C++, SQL

Application Software: Google Colab, Jupyter Notebook, Git, Github, Power BI 2, Tableau 2023.2.0, R Studio, Microsoft Excel

Databases: SQL (MySQL, PostgreSQL pgAdmin 4 v8), NoSQL (MongoDB)

Certifications & Training: DeepLearning.AI Data Engineering Professional Certificate, Linux Professional Training (IIT Bombay)

WORK EXPERIENCE

PROJECT 990

Bloomington, IN, USA

Data Engineering Intern

May 2025 – Present

- Designed and deployed a robust Python-based data migration pipeline that transformed unstructured IRS tax filings into structured, queryable datasets—enhancing data accessibility, reliability, and downstream analytics efficiency.
- Queried and examined 2 million+ IRS tax records using advanced SQL techniques, uncovering critical data anomalies, schema mismatches, and reporting gaps, while spearheading initiatives for continuous data-quality improvements and audit readiness.
- Defined, tracked, and enforced data-quality KPIs to keep ingested and processed data in sync, giving stakeholders the confidence to make sound, data-driven decisions.

TECHNICAL CODING RESEARCH INNOVATION

Mumbai, MH, India

Data Science & Machine Learning with Python Intern

January 2022 – April 2022

- Performed employee retention analysis using a structured ETL pipeline, ensuring accurate data ingestion from over 2900 records.
- Built classification models in Python, achieving 85.26% test accuracy to identify high-risk attrition cases.
- Created data visualizations (bar charts, heatmaps) aligned with HR metrics, influencing workforce planning decisions.

PROJECTS

TWITTER DATA PIPELINE WITH APACHE AIRFLOW AND AWS

September 2025

- Built an end-to-end Twitter → S3 data pipeline (Python, Tweepy, pandas, s3fs) orchestrated with Apache Airflow on Ubuntu EC2; scheduled hourly to ingest ≈1000 tweets/run, transform to normalized CSV, and land to S3 with <3 min end-to-end latency per run.
- Productionized with Airflow DAGs (3 Python tasks: extract/transform/load) using retries & logging; achieved ~99% task success over 50+ scheduled runs, generating 24 versioned objects/day in S3 and reducing manual collection effort by ~100%.
- Hardened and automated AWS setup: provisioned EC2, configured IAM role-based S3 access (no embedded credentials), corrected SSH key/file permissions, and exposed the Airflow UI on 8080; enabled zero manual steps for recurring runs and 100% scheduled execution once deployed.

YOUTREND DATA LAKE

August 2025

- Designed and deployed a data lake on AWS S3 to store and manage over 100,000 records of trending YouTube video data, implementing robust partitioning and cataloging via AWS Glue, resulting in 99% data query success rate for downstream analytics.
- Automated and optimized ETL pipelines using AWS Lambda and Glue, reducing data processing time by 60% and enabling daily (vs. weekly) refresh of cleaned, query-ready datasets for use in analytics dashboards.
- Built interactive dashboards by integrating AWS Athena and Power BI, supporting real-time analysis of video popularity metrics and enabling actionable business insights 4x faster than prior manual spreadsheet workflows.

GLOBAL SUPERSTORE ANALYTICS – AWS END-TO-END DATA PIPELINE AND DASHBOARD

July 2025

- Designed and implemented a scalable data pipeline on AWS (S3, Glue Crawler + ETL, Redshift) to process Global Superstore data, improving overall data usability by ~10% and reducing ETL runtime by ~5% through schema optimization and structured transformations.
- Queried and analyzed transformed datasets with Athena and built an interactive Power BI dashboard (10+ KPIs, sales trends, profitability insights), helping reduce manual reporting effort by ~15% and making insights more accessible to business stakeholders.
- Configured IAM roles, resolved connectivity/authentication issues, and implemented CloudWatch monitoring, improving pipeline stability and reducing troubleshooting time by ~10%.